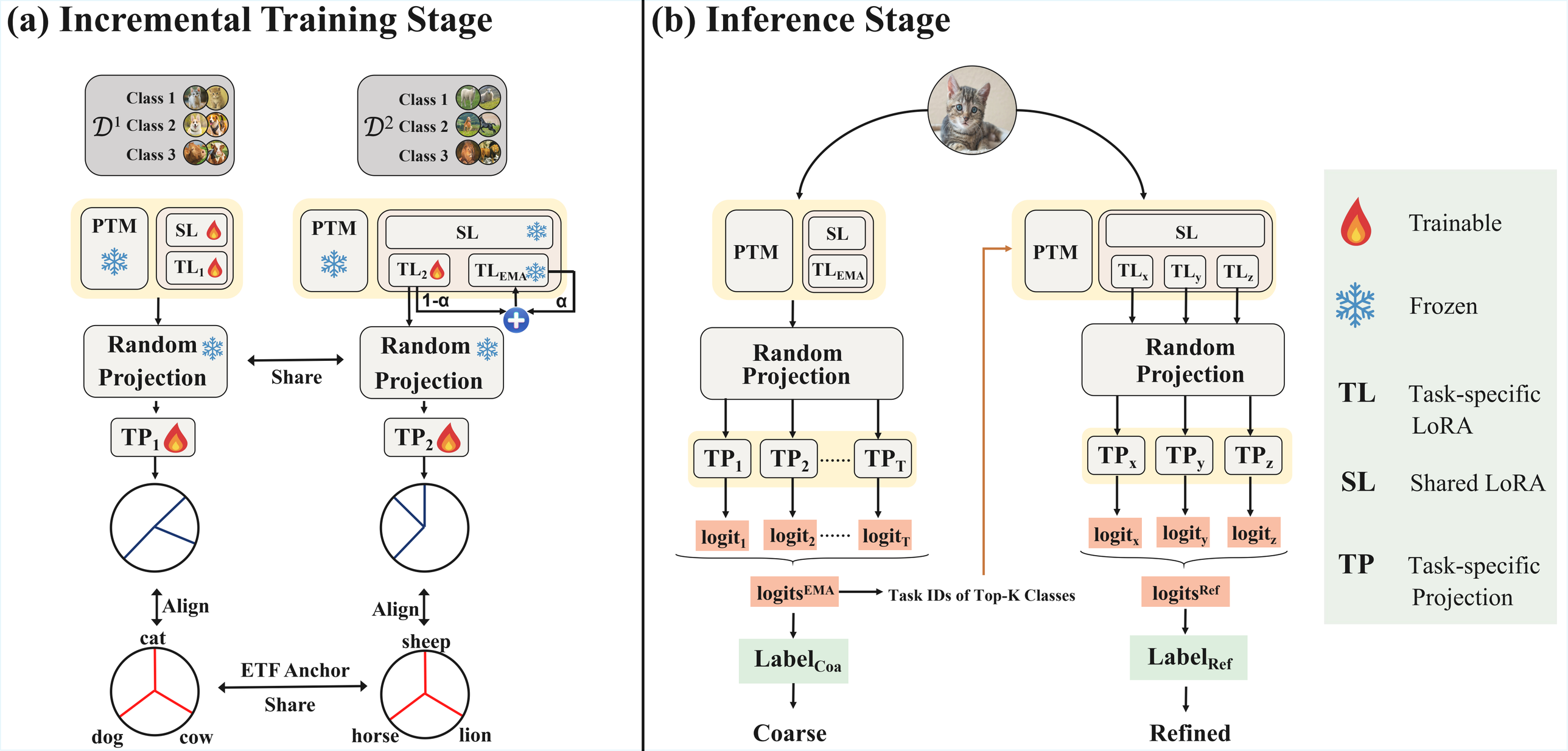
(a) Incremental Training Stage
Class 1
$\mathcal{D}^1$ Class 2
Class 3
Class 1
$\mathcal{D}^2$ Class 2
Class 3
PTM
SL
TL_1
PTM
SL
TL_2
TL_{EMA}
1-α
α
Random Projection
Share
Random Projection
TP_1
TP_2
Align
Align
cat
sheep
ETF Anchor
Share
dog
cow
horse
lion
(b) Inference Stage
PTM
SL
TL_{EMA}
PTM
SL
TL_x
TL_y
TL_z
Random Projection
Random Projection
TP_1
TP_2
TP_T
TP_x
TP_y
TP_z
$logit_1$
$logit_2$
$logit_T$
$logit_x$
$logit_y$
$logit_z$
$logits^{EMA}$
Task IDs of Top-K Classes
$logits^{Ref}$
$Label_{Coa}$
$Label_{Ref}$
Coarse
Refined
Trainable
Frozen
TL Task-specific LoRA
SL Shared LoRA
TP Task-specific Projection